

Emmy went on to develop a whole new type of mathematics called “abstract algebra,” which is like a special language that mathematicians and scientists use to explore, describe, and understand the universe better.

In the 1930s, a group called the Nazis ruled Germany, and it became a very dangerous place for Jewish people. Emmy decided to move to the United States. She quickly became as respected there as she had been in Europe. The tools she developed have been used by scientists all over the world, helping to explain everything from the behavior of tiny particles to the gravity of black holes.

Emmy's amazing math helps us to understand the universe.

Mathematical mind

At first, Emmy had to work in roles without pay. But, before long, her incredible mathematical talent meant that lots of leading scientists and mathematicians wanted to work with her.

Emmy helped to solve a puzzle created by Albert Einstein's new theory of gravity. She proved two mathematical theorems, or rules, that are now woven into all kinds of physics. Many people, including Albert himself, demanded that Emmy be paid for her work. In 1922, she was finally given a role as a college researcher and professor.

Emmy loved to chat about algebra, especially during long walks and picnics with her students.